

KADENA AB, Japan

WMO: 479310

Lat: 26.356N

Lon: 127.768E

Elev: 153

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 42204

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 48.0	(c) 50.4	(d) 37.1	(e) 32.6	(f) 55.2	(g) 39.0	(h) 35.2	(i) 56.1	(j) 24.5	(k) 60.0	(l) 22.8	(m) 60.1	(n) 6.7	(o) 60	(p) 0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	91.5	80.4	90.0	80.5	89.3	80.5	84.8	86.9	83.6	86.3	82.7	85.6	12.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.4	181.7	85.9	83.0	173.2	84.9	82.2	168.5	84.4	49.4	86.8	48.0	86.8	46.9	85.2	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.5	(b) 22.5	(c) 19.9	(e) 42.8	(f) 93.0	(g) 3.0	(h) 1.9	(i) 40.6	(j) 94.3	(k) 38.8	(l) 95.4	(m) 37.1	(n) 96.4	(o) 34.9	(p) 97.8		
(a) 26.5	(b) 22.5	(c) 19.9	(e) 40.7	(f) 85.0	(g) 2.1	(h) 2.0	(i) 39.1	(j) 86.5	(k) 37.9	(l) 87.7	(m) 36.7	(n) 88.8	(o) 35.2	(p) 90.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	61.3	62.2	64.9	69.8	75.0	80.8	84.4	84.0	81.8	76.9	71.1	64.4
	DBStd	9.21	4.71	5.10	5.43	4.97	4.09	3.68	2.04	2.15	2.83	3.83	4.40	4.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	405	135	109	73	53	0	0	0	0	0	0	5	70
	CDD50	8432	351	341	460	595	775	923	1065	1054	953	835	633	447
	CDD65	3362	21	30	68	158	310	473	600	589	503	370	188	52
	CDH74	31449	19	36	126	590	1924	4832	7445	7253	5519	2839	776	90
	CDH80	10504	0	0	5	56	286	1489	3125	2985	1884	613	59	2
(14) Wind	WSAvg	10.5	10.3	10.4	10.4	10.1	9.7	11.0	10.9	10.6	10.6	11.0	10.5	10.4
(15) Precipitation	PrecAvg	81.8	4.8	4.3	6.0	6.5	9.6	10.1	7.5	9.8	7.1	6.3	4.3	5.5
	PrecMax	123.4	12.2	16.0	14.4	18.1	27.2	22.4	24.4	25.6	18.1	23.0	10.2	32.4
	PrecMin	40.7	1.1	1.1	0.8	0.8	0.6	0.9	0.5	1.5	1.7	0.2	0.2	0.9
	PrecStd	21.4	2.2	3.3	3.5	3.9	5.5	5.5	6.7	5.8	3.8	4.8	3.0	5.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	77.3	80.4	84.1	86.4	91.1	93.3	93.4	91.6	89.5	84.0	79.1
		MCWB	67.6	69.5	72.3	76.6	79.0	80.5	80.8	80.0	78.4	76.8	75.3	70.7
	2%	DB	73.6	75.0	77.1	82.0	84.6	89.2	91.5	91.5	89.8	86.8	82.0	76.7
		MCWB	66.0	68.4	71.3	75.2	78.1	81.2	80.7	80.8	79.3	78.0	74.0	69.2
	5%	DB	71.7	73.0	75.1	79.1	82.8	87.8	89.9	89.9	88.2	84.9	80.2	74.6
		MCWB	64.8	67.3	69.2	73.0	77.1	81.5	81.0	80.7	79.6	77.1	72.9	67.4
	10%	DB	69.8	71.2	73.3	77.2	81.0	86.3	89.2	88.9	87.4	83.4	78.5	72.8
		MCWB	63.4	65.8	68.0	72.1	75.8	80.7	81.0	80.6	79.5	75.9	71.8	66.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	72.6	74.6	79.1	82.8	86.3	86.3	86.3	85.6	84.0	78.9	74.1
		MCDB	73.8	75.5	77.2	81.2	83.8	87.3	88.0	88.2	86.8	84.8	81.3	76.6
	2%	WB	68.7	70.6	72.6	76.9	80.4	83.9	84.4	84.3	83.6	81.0	77.0	71.5
		MCDB	71.5	73.2	75.5	79.5	82.4	85.8	87.3	87.2	85.7	82.9	79.0	74.3
	5%	WB	66.5	68.8	71.1	75.1	78.8	82.5	83.3	83.0	82.1	79.2	75.3	69.4
		MCDB	70.2	71.4	74.0	78.2	81.5	84.9	86.5	86.2	85.0	82.7	78.0	72.5
	10%	WB	64.4	66.8	69.3	73.6	77.6	81.3	82.4	82.1	80.8	77.9	73.5	67.4
		MCDB	68.4	69.9	72.3	76.6	80.3	84.5	85.9	85.6	84.5	81.8	76.9	71.3
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	11.9	12.2	11.6	10.4	8.7	9.3	9.6	10.4	11.4	12.0	12.1
		MCDBR	14.4	13.4	12.4	11.4	10.6	9.1	10.7	11.0	11.2	12.2	12.8	14.0
		MCWBR	9.7	9.2	8.9	7.1	6.5	4.9	4.5	4.8	5.6	6.5	7.5	8.8
	5% WB	MCDBR	11.1	10.9	10.8	9.8	9.1	8.4	9.0	9.3	9.3	9.8	9.5	10.9
		MCWBR	9.4	9.2	9.4	7.3	6.9	5.8	5.1	5.5	6.1	6.5	7.2	8.9
(40) Clear-Sky Solar Irradiance	taub		0.458	0.487	0.560	0.568	0.513	0.471	0.471	0.488	0.484	0.482	0.465	0.457
	taud		2.131	2.068	1.899	1.939	2.161	2.343	2.339	2.278	2.259	2.204	2.207	2.141
	Ebn at Noon		246	249	238	240	253	262	262	257	256	248	243	240
	Edh at Noon		43	49	62	61	49	40	40	43	43	43	40	41
(44) All-Sky Solar Radiation	RadAvg		765	964	1195	1398	1554	1718	2064	1886	1652	1295	944	772
	RadStd		91	146	114	135	202	156	168	127	123	111	87	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.84	N/A	N/A	+0.92	+1.13	N/A	+90	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.74	+0.95	N/A	N/A	N/A	N/A

Nomenclature: See separate page